

RAG And LangChain

LangChain

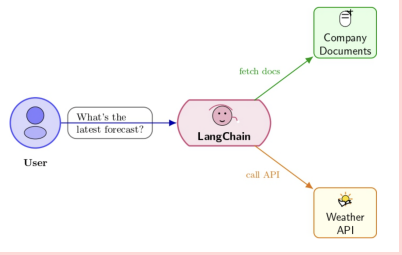
1 What is LangChain?

2 Key Concepts in LangChain

3 LangChain Ecosystem

4 How It Works

LangChain is an open-source framework designed to build applications with large language models (LLMs) by combining them with external data, logic, and memory.



LangChain is not an LLM itself, it's a framework that orchestrates LLMs and external components.

LLMs – Large language models you want to use (OpenAI, Gemini, Llama, GPT, etc.).

Prompts – Templates that guide the LLM on how to respond.

Chains – Sequences of steps that transform input into output using LLMs and tools.

Agents – LLM-powered programs that decide which tool to use next to solve a problem.

Memory – Keeps track of previous interactions for multi-turn conversations.

Vector Stores / Retrieval – Databases that store embeddings for retrieval-augmented generation (RAG).

Callbacks – Hooks to track and log what the model is doing.

LangChain is more like a full-stack ecosystem for building intelligent apps. Components can be combined depending on your use case.

LLMs and Chat Models

Prompts and Prompt Templates

Chains

Agents

Memory

Document Loaders and Transformers

Vector Stores

Tools and APIs

Callbacks

5 Example: How LangChain fits together

1 Document Loader → Load PDFs into memory

2 Text Splitter → Split into smaller chunks

3 Embeddings → Convert chunks to vector embeddings

4 Vector Store → Store embeddings for semantic search

5 Retriever → Retrieve top-k relevant chunks for a query

6 LLM / Chain → Feed query + retrieved chunks to LLM

7 Answer → Return generated answer

6 Why LangChain is Powerful

1 Abstracts away the complexity of integrating LLMs + tools + memory + retrieval

2 Modular: use only what you need

3 Allows multi-step reasoning and tool use

4 Works with any LLM or local model

5 Enables building production-ready AI applications

7 Let's Go Depth Deeper Into It

Introduction to LangChain

LangChain Components

LangChain Models

Prompts in LangChain

Structured Output in LangChain

Output Parsers in LangChain

Chains in LangChain

Runnables in LangChain

Document Loaders in LangChain

Text Splitters in LangChain

Vector Stores in LangChain

Retrievers in LangChain

Retrieval Augmented Generation (RAG)

Code

<https://codeshare.io/aJ6rnE>

<https://codeshare.io/aJ6rnE>

RAG

What is Retrieval Augmented Generation (RAG)?

Retrieval Augmented Generation (RAG) is a technique that enhances LLMs by integrating them with external data sources.

RAG enables AI systems to produce more accurate and contextually relevant responses.

LLMs are powerful but come with inherent limitations:

Limited knowledge

Hallucinations

Generic responses

How Does RAG Work?

Step 1: Data collection

Step 2: Data chunking

Step 3: Document embeddings

Step 4: Handling user queries

Step 5: Generating responses with an LLM

Scraping

Store(Chroma , PineCone)

Fusion-in-decoder (FiD) and Fusion-in-Encoder (FIE)

Overview



Code

<https://codeshare.io/5DVVWVr>